

Introduction to data summarizing and visualization

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The R programming language provides many tools for data analysis and visualization, but all the options can be daunting. This lesson provides an introduction to wrangling data in R and using graphics to tell a story with data.

Learning objectives

1. Understand the difference between files and R objects
 2. Modify data for proper data hygiene
 3. Summarize information from raw data
 4. Visualize data to convey information
-

Getting started

The tools: R and RStudio

For this lesson, we will use the R programming language in the RStudio environment. RStudio provides a convenient interface for working with files and packages in R. If you have not done so already, install R and RStudio; details can be found on the [installation page](#).

Preparing our workplace

Key to successful programming is organization. In RStudio, we use Projects to organize our work (a “Project” is really just a fancy name for a folder that contains all our files). For this lesson, we’ll create a new project through the File menu (File > New Project). In the first dialog, select “New Directory”, and select “New Project” in the second dialog. Next you’ll be prompted to provide a directory name. This will be the name of our project, so we should give it an informative name. For this lesson, we will be using data from vegetation surveys of [Tumacacori National Historical Park](#); so give our directory a descriptive name, like “vegetation”. We need also to tell RStudio where to put the lesson on our computer; for this lesson, we will place the folder on our Desktop, so it is easy to find. In your own work, you may find it better to place project folders in your Documents folder.

The last thing we need to do to set up our workspace is to use file organization that reinforces best practices. In general, there should be a one-way flow of information: we take information from *data* and write code to produce *output*. We want to avoid any output from messing up our data, so we create separate folders for each. We want to create two folders, one for our data and one for any output, which may include results of statistical analyses or data visualization. In the R console,

```
dir.create("data")
dir.create("output")
```

Getting the data

We are going to use vegetation survey data from the Tumacacori National Historical Park. These data are based on several surveys and are available through the National Park Service’s [Integrated Resource Management Applications Data Store](#).

We can use R to directly download the data from the web. In the R console, enter:

```
download.file(url = "https://tinyurl.com/tnhp-data",
              destfile = "data/tumacacori-vegetation.csv")
```

(You can also download the file manually by entering <https://tinyurl.com/tnhp-data> into a web browser’s address bar.)

Note if you click on the data folder in your project (there should be a file browser in the lower-right corner of your RStudio window), you should see a single file, “tumacacori-vegetation.csv”, that shows up as 31.5 KB. If you do not see that file, or if the file size is listed as 0 KB, try downloading the file again, making sure you spelled the URL and destination file name correctly.

Data in R

Data *outside* R

We are going to start by looking at those data we downloaded in a spreadsheet program like Excel or [LibreOffice](#). Open your spreadsheet program and open the file we just downloaded. We can see the data are organized in columns and rows. This is a good example of what is known as “tidy data” where there is one observation per row and one type of data per column. In these data, each row corresponds to a single plant species in a single plot, and the columns have different data about that species, including the family name and the percent cover in that plot.

Take a moment to look at those data but don’t change any of the values in the cells. Close the file, and make sure *not* to save any changes to the file.

Data in R

We are going to work with those same data in R. To do so we will need to read the data in to R’s memory, but first we are going to start a script so all the steps we take for data analyses are going to be saved in one place. Make a new script through the File menu, via File > New File... > R Script. We’ll start by adding a little bit of information at the beginning of the script:

```
# Plotting Tumacacori vegetation
# Jeffrey C. Oliver
# jcoliver@email.arizona.edu
# 2019-09-06
```

Note that RStudio is also telling us that we have unsaved changes: the text in the tab is red and there is a little asterisk up by the file name (which is probably “Untitled1”, which should also be a clue). We should save this file via Ctrl-S (Window or Linux) or Cmd-S (MacOS), giving it a short but descriptive name. We’ll use `tumacacori-plots.R`.

Now we are finally going to read data into R. When you opened the file in Excel you might have noticed that it has the file extension “csv”, which stands for Comma-Separated Values. The CSV file format is common for data tables and has the benefit that it can be easily read by many programs, which is not always the case for .xls and .xlsx files. To read the data into R, we use the `read.csv` function:

```
plant_data <- read.csv(file = "data/tumacacori-vegetation.csv",
                       stringsAsFactors = TRUE)
```

The statement shows typical syntax for an R command. From an abstract perspective, most R statements look like:

$$\text{Variable Name} \leftarrow \text{Function Name}(\text{Function Arguments})$$

The *function* (`read.csv`) is given two pieces of information, or *arguments*:

1. `file` which tells R where the file is located (in this case, in the data folder)
2. `stringsAsFactors` which tells R to treat text data as categorical data

The data in the file are read into R, then stored in a *variable* called `plant_data`.

Quality Assurance

Whenever we start writing a new script, we *always* want to make sure the data are being read in correctly. When we are doing this initial quality check, we don't necessarily need to record all the commands we type, so we can use the R console to type commands. We'll start with `head` which shows the first six rows of data:

```
head(plant_data)
```

	Plot_Code	Field_Name	Common_Name	Family
1	TUMC_IP01	Acacia constricta	whitethorn acacia	Fabaceae
2	TUMC_IP01	Acacia greggii	catclaw acacia	Fabaceae
3	TUMC_IP01	Amaranthus palmeri	carelessweed	Amaranthaceae
4	TUMC_IP01	Aristida ternipes	spidergrass	Poaceae
5	TUMC_IP01	Bidens leptcephala	fewflower beggarticks	Asteraceae
6	TUMC_IP01	Boerhavia spicata	creeping spiderling	Nyctaginaceae

	Species_Code	Percent_Cover	Leaf_Type	Leaf_Phenology	Community
1	ACACON	30.0	Microphyllous	Drought-deciduous	Shrubland
2	ACAGRE	30.0	Microphyllous	Drought-deciduous	Shrubland
3	AMAPAL	3.0	Microphyllous	Drought-deciduous	Shrubland
4	ARITER	0.5	Microphyllous	Drought-deciduous	Shrubland
5	BIDLEP	18.0	Microphyllous	Drought-deciduous	Shrubland
6	BOESPI	8.0	Microphyllous	Drought-deciduous	Shrubland

Similarly, the `tail` function shows the last six rows of data:

```
tail(plant_data)
```

	Plot_Code	Field_Name	Common_Name	Family
286	TUMC_IPS4	Eragrostis lehmanniana	Lehmann lovegrass	Poaceae
287	TUMC_IPS4	Erioneuron avenaceum	shortleaf woollygrass	Poaceae
288	TUMC_IPS4	Opuntia engelmannii	cactus apple	Cactaceae
289	TUMC_IPS4	Prosopis velutina	velvet mesquite	Fabaceae
290	TUMC_IPS4	Salsola kali	Russian thistle	Chenopodiaceae
291	TUMC_IPS4	<NA>	Unknown Perennial Forb	<NA>

	Species_Code	Percent_Cover	Leaf_Type	Leaf_Phenology	Community
286	ERALEH	0.5	Microphyllous	Drought-deciduous	Shrubland
287	ERIAVE	0.5	Microphyllous	Drought-deciduous	Shrubland
288	OPUENG	0.5	Microphyllous	Drought-deciduous	Shrubland
289	PROVEL	50.5	Microphyllous	Drought-deciduous	Shrubland
290	SALKAL	0.5	Microphyllous	Drought-deciduous	Shrubland
291	UNKFOR	8.0	Microphyllous	Drought-deciduous	Shrubland

Note in the output of `tail` there are rows with `<NA>` values. In R, NA has a special meaning: it indicates missing values in the data. We'll deal with that in a bit, but remember that missing data may require some special handling in R.

One more useful function is `str`, which stands for “structure”:

```
str(plant_data)
```

```
'data.frame':  291 obs. of  9 variables:
 $ Plot_Code      : Factor w/ 22 levels "TUMC_IP01","TUMC_IP02",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ Field_Name     : Factor w/ 71 levels "Acacia constricta",...: 1 2 5 9 12 14 18 37 44 45 ...
 $ Common_Name    : Factor w/ 74 levels "Annual Forb",...: 73 15 14 57 31 20 55 12 72 16 ...
 $ Family         : Factor w/ 26 levels "Acanthaceae",...: 12 12 3 20 4 17 20 7 15 12 ...
 $ Species_Code   : Factor w/ 74 levels "ACACON","ACAGRE",...: 1 2 7 11 14 16 19 39 46 47 ...
 $ Percent_Cover  : num  30 30 3 0.5 18 8 18 0.5 0.5 3 ...
 $ Leaf_Type      : Factor w/ 3 levels "Broad-leaved",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ Leaf_Phenology: Factor w/ 3 levels "Cold-deciduous",...: 2 2 2 2 2 2 2 2 2 2 ...
 $ Community      : Factor w/ 5 levels "Forest","Shrubland",...: 2 2 2 2 2 2 2 2 2 2 ...
```

The output of `str` shows the size of the data, in terms of number of rows and columns, and the type of data in each column.

Cleaning up

Rarely are data “ready to go” when they are read into R. The data we are using are going to require two adjustments: removal of rows with missing data and a selection of only a portion of the data.

Missing data

To remove rows that have missing data, we use the `na.omit` function. Before we do, though, look at the “Environment” tab of your workspace. There should be a row for `plant_data` that indicates the size of our data set. In this case, it should show “291 obs. of 9 variables”, indicating we have 291 rows of data, with 9 columns. Now let’s drop those rows with missing data, putting this command in the script file, not the R console:

```
plant_data <- na.omit(plant_data)
```

Look again at the Environment tab. There should only be 287 obs. because we removed those rows with missing data. The number of columns should not change at all. One thing we are going to do now is to start commenting our code. We use the pound or hash sign (`#`) to tell R that we are writing a comment that should not be interpreted as code. For that line we just wrote, we are going to add a little note about *why* we did what we did.

```
# Do not want rows with missing data  
plant_data <- na.omit(plant_data)
```

Subsetting data

The dataset include observations of 26 different plant families at the plots. For our purposes, we are going to focus on a few of the families that make up most of these communities, namely the legumes (Fabaceae), grasses (Poaceae), mustards (Brassicaceae), and amaranths (Amaranthaceae).

```
# Only focus on four families  
families_keep <- c("Fabaceae", "Poaceae", "Brassicaceae", "Amaranthaceae")  
  
# Create new data with only four families  
subset_data <- plant_data[plant_data$Family %in% families_keep, ]  
  
# Re-level data in the Family column  
subset_data$Family <- factor(subset_data$Family)
```

In those three lines, we:

- Created a list of the names of the families we are interested in,
- Made a new variable called `subset_data` and stored the data for only those four families, and
- “Re-leveled” the data in the Family column (no need to worry too much about what this means now, it will just make plotting easier later on).

Look again at your Environment tab, you should see another item listed, this one called `subset_data`. It *should* have 164 rows (observations). If it doesn’t, check your work to make sure you spelled all the family names correctly.

What about that spreadsheet file?

So we’ve done some manipulations to the data, dropping rows with missing data and creating a subset for four families. Did that do anything to the original spreadsheet file? Open the file in your spreadsheet program (like Excel or LibreOffice) and take a look. If you look towards the bottom, you can see that those rows with NA are still there. Compare this with the output of `tail(plant_data)`. Note the `plant_data` in R does not have the rows with NA values because we removed them with `na.omit` above. This demonstrates that modifying data in R *does not change the data in the original data files*. Files are only changed when we explicitly tell R to write changes to the hard drive. Since we do not want those changes written to our original data file, we are *not* going to have R write any data to the files.

Summarizing data

We would like to get some summary data from our data, and R provides functions for making this easy.

Using summary

The `summary` function works on data frames (which is how R stores spreadsheets). It provides a little bit of information about each of the columns of data. For columns that have categorical data, like `Common_Name`, it tells us how many rows have each category. For columns with numerical data, it provides some statistics, such as the mean, median, minimum, and maximum.

```
summary(subset_data)
```

Plot_Code	Field_Name	Common_Name
TUMC_IP12: 13	Prosopis velutina :21	velvet mesquite :21
TUMC_IP15: 11	Acacia greggii :19	catclaw acacia :19
TUMC_IP05: 10	Amaranthus palmeri :17	carelessweed :17
TUMC_IP14: 10	Bouteloua curtipendula :14	sideoats grama :14
TUMC_IP20: 10	Descurainia pinnata :14	western tansymustard:14
TUMC_IP02: 9	Setaria :12	bristlegrass :12
(Other) :101	(Other) :67	(Other) :67

Family	Species_Code	Percent_Cover	Leaf_Type
Amaranthaceae:17	PROVEL :21	Min. : 0.50	Broad-leaved : 11
Brassicaceae :18	ACAGRE :19	1st Qu.: 0.50	Broad-leaved herbaceous: 4
Fabaceae :53	AMAPAL :17	Median : 3.00	Microphyllous :149
Poaceae :76	BOUCUR :14	Mean :12.73	
	DESPIN :14	3rd Qu.:18.00	
	SETAR :12	Max. :86.00	
	(Other):67		

Leaf_Phenology	Community
Cold-deciduous : 11	Forest :15
Drought-deciduous:149	Shrubland :70
Herb - annual : 4	Wooded Herbaceous: 4
	Wooded Shrubland :71
	Woodland : 4

Looking at this output, we see, for instance, there are 21 rows with data for velvet mesquite. We also see the mean percent cover is 12.73 across all plots and species.

Summary statistics for groups

But what if we wanted information about certain groups of data? For example we might want the mean and standard deviation of percent cover for a particular species. We can calculate those statistics for all species with the `mean` and `sd` functions:

```
# Extract summary statistics for all species in subsetted data
cover_mean <- mean(subset_data$Percent_Cover)
cover_sd <- sd(subset_data$Percent_Cover)
```

We can see the values of each of these by typing in the variable name alone into the R console:


```
cover_mean
```

```
[1] 12.72561
```

```
cover_sd
```

```
[1] 16.48287
```

But how do we get these statistics for a single species, say velvet mesquite? We can perform a data subset *within* the calls to `mean` and `sd`:

```
# Calculate mean and standard deviation for mesquite alone
mesquite_mean <- mean(subset_data$Percent_Cover[subset_data$Common_Name == "velvet mesquite"])
mesquite_sd <- sd(subset_data$Percent_Cover[subset_data$Common_Name == "velvet mesquite"])
```

In the commands above, we still used `mean` and `sd` on the `Percent_Cover` column but here we *only* used those rows where the value in the `Common_Name` column was “velvet mesquite”. Typing the variable names into the R console will show us the statistics for velvet mesquite.

```
mesquite_mean
```

```
[1] 27.07143
```

```
mesquite_sd
```

```
[1] 21.14077
```

Note that R is case-sensitive, meaning that lower-case letters and upper-case letters are different in R. If we changed the calculation for the mean from

```
mesquite_mean <- mean(subset_data$Percent_Cover[subset_data$Common_Name == "velvet mesquite"])
```

to (capitalizing the “V” and the “M”):

```
mesquite_mean <- mean(subset_data$Percent_Cover[subset_data$Common_Name == "Velvet Mesquite"])
```

the result will be `NaN` which stands for “Not a Number”. This is because there are zero rows with the common name of “Velvet Mesquite”, and the mean of nothing is not definable.

What if you wanted means and standard deviations for each species? You could copy and paste the code, changing the variable name and species name for every species, but since there are 32 different common names in these data, that would be tedious and a great way to make a mistake. Thankfully, there are additional packages in R that make this much easier.

Summary statistics made easier

To get these summary statistics for all species, we are going to use a third-party package. What does that mean, “third-party”? When we think of software, there are generally two parties: the one that wrote the software and the one that uses the software. In this case, the R Core Team wrote the R software and we are the second party, the users of R. Third-party packages are those written by someone other than the authors of the original software. R is especially amenable to third-party development and some of the most widely used packages in R were developed by teams other than the R Core Team. Importantly, if we are using third-party R packages, we need to take two steps: first we need to **install** the package, second we need to **load** the package’s functions into memory. The first (installation) only has to happen once on your machine; the second (loading into memory) has to happen every time you use R.

We are going to use the `dplyr` package for summarizing our data. To install `dplyr`:

```
install.packages("dplyr")
```

Now that the package is installed, we can load it into memory with the `library` command. When using third-party packages, we generally add `library` commands to the start of the script, so anyone reading our script can tell which, if any, additional packages the script requires to run.

```
library("dplyr")
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

Now we can use `dplyr` functions to generate summary statistics for *each* species. Here we use the `group_by` function to essentially create a separate pile of data for each species, then, for each “pile”, calculate the mean and standard deviation:

```
# Calculate mean and standard deviation for each species separately
summary_stats <- subset_data %>%
  group_by(Common_Name) %>%
  summarize(mean_cover = mean(Percent_Cover),
            sd_cover = sd(Percent_Cover))
```

If we consider the above code using English, it is easier to understand if we replace all the pipes (%>%) with the word “then”:

```
summary_stats <- subset_data, then
  group_by(Common_Name), then
  summarize(mean_cover = mean(Percent_Cover),
            sd_cover = sd(Percent_Cover))
```

Or, using words only,

```
Make a new variable called "summary_stats", take the subset_data, then
  make a separate group of data for each Common_Name, then
  calculate the mean and standard deviation of Percent_Cover and store them
    in columns called "mean_cover" and "sd_cover" respectively.
```

We can also add this explanation to our code through the use of comments:

```
# Calculate mean and standard deviation for each species separately
summary_stats <- subset_data %>%
  # make a separate group of data for each Common_Name
  group_by(Common_Name) %>%
  # calculate the mean and standard deviation of Percent_Cover and store them
  #   in columns called "mean_cover" and "sd_cover" respectively
  summarize(mean_cover = mean(Percent_Cover),
            sd_cover = sd(Percent_Cover))
```

We can see these summary stats by typing the variable name, `summary_stats` into the R console:

```
summary_stats
```

```
# A tibble: 32 x 3
  Common_Name      mean_cover sd_cover
  <fct>          <dbl>    <dbl>
1 Bermudagrass      8        0
2 big sacaton     13.6     21.7
3 bristlegrass     1.54     1.29
4 broomcorn millet  3         NA
5 bush muhly      14.3     11.3
6 cane bluestem    0.5         NA
7 carelessweed    20.9     18.7
8 catclaw acacia   19.2     17.5
9 catclaw mimosa   1.75     1.77
10 cliff muhly     13        7.07
# i 22 more rows
```

Get % cover for each family/plot

Ultimately, we would like to look at how much cover there is for each plant family in the Tumacacori plots. Right now the data are shown for each *species*, but we would like to know how much of the plant cover corresponds to each *family*. We'll use the dplyr package again for summarizing data, but this time we are going to group by several levels: first by the plot, then by family, then by the community.

```
# Calculate family-level total percent cover for each plot separately
family_data <- subset_data %>%
  group_by(Plot_Code, Family, Community) %>%
  summarize(Family_Percent_Cover = sum(Percent_Cover))
```

``summarise()`` has grouped output by 'Plot_Code', 'Family'. You can override using the ``groups`` argument.

Once again, thinking about this code in English, we can replace all the pipes (`%>%`) with the word “then”:

```
Make a new variable called family_data, take the subset_data, then
  make a separate group of data for each Plot_Code, break those groups up
    into smaller groups, one for each Family, divide those groups into separate
      groups for Community, then
add up all the values in the Percent_Cover column and store it
  in a column called "Family_Percent_Cover"
```

If we look at these data, we can see we now have total percent cover for each family:

```
head(family_data)
```

```
# A tibble: 6 x 4
# Groups:   Plot_Code, Family [6]
  Plot_Code Family      Community      Family_Percent_Cover
  <fct>      <fct>      <fct>              <dbl>
1 TUMC_IP01 Amaranthaceae Shrubland           3
2 TUMC_IP01 Fabaceae      Shrubland          71.5
3 TUMC_IP01 Poaceae       Shrubland          48
4 TUMC_IP02 Amaranthaceae Wooded Shrubland    63
5 TUMC_IP02 Brassicaceae Wooded Shrubland     0.5
6 TUMC_IP02 Fabaceae      Wooded Shrubland    14.5
```

Visualizing data

When we are using data to tell a story, often we use a visualization in order to make a point. In this case, we are especially interested in looking at how the percent cover of different families of plants differs (or doesn't differ) among the different plant communities. For example, we would be interested to know if the percent cover of the Fabaceae, which includes mesquites and acacias, differs between those plots categorized as forest and those plots categorized as shrubland.

First rule of plots

Whenever we want to visualize data, it is almost best to draw it out by hand. This makes it easy to evaluate different graphical representations of data. So to start, take a moment to consider the data in `family_data` and how it could be visualized. Keep in mind the question above about percent cover differences among plant communities. Don't worry about exact values, just think about how a drawing could illustrate the point.

The ggplot2 package

In order to visualize the data in R, we are going to use the ggplot2 package. Like we did for the dplyr package, we will first need to install the package:

```
install.packages("ggplot2")
```

As we did with the dplyr package, we use the `library` function to load the functions into memory (remember, this should go at the beginning of your script):

```
library("ggplot2")
```

Code it second

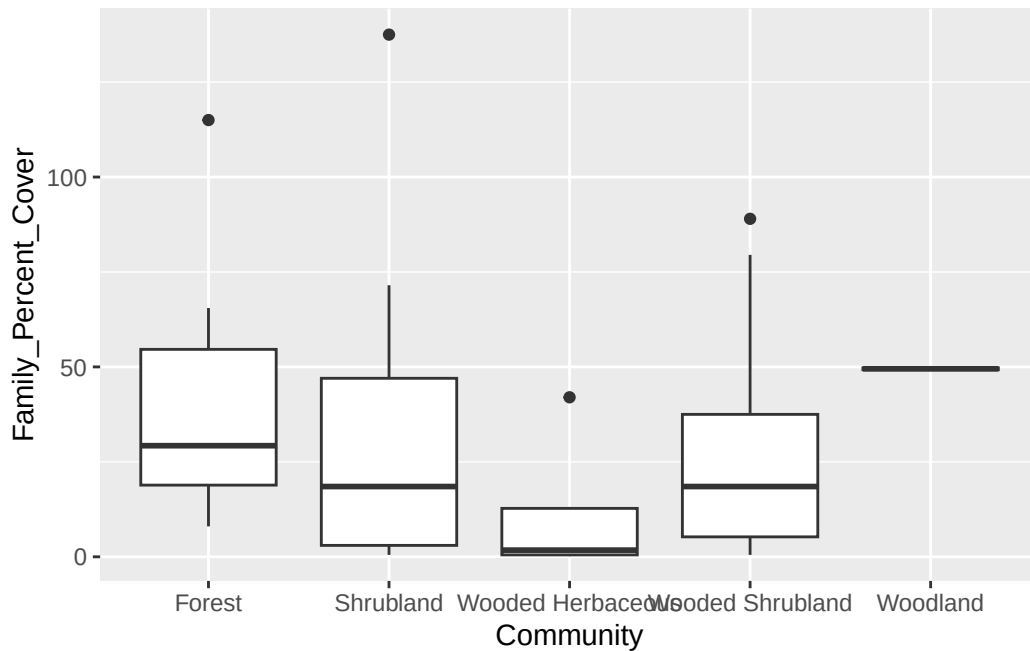
Now that we have the ggplot2 package installed, we are going to make a boxplot of the data. Boxplots are especially good for comparing values of different categories. Here we are going to create a boxplot for percent cover, with different communities shown on the x-axis. So here we go.

```
# Boxplot of family-level data
cover_plot <- ggplot(data = family_data,
                     mapping = aes(x = Community, y = Family_Percent_Cover)) +
  geom_boxplot()
print(cover_plot)
```

Let's break down that code:

- `ggplot` is the main function for creating graphics in the ggplot2 package, we start by passing it two pieces of information:
 - `data` tells R which data to use, in this case we are using the percent cover sums we stored in `family_data`
 - we use the `mapping` argument to tell R which variable to put on the x-axis (`Community`) and which variable to put on the y-axis (`Family_Percent_Cover`)
- We also have to tell R *how* to plot the data. Because we are interested in creating a boxplot, we use the command `geom_boxplot`.
- Finally, we tell R to draw the plot with the `print` command. That is, we created an entire plot object and stored it in the variable `cover_plot`. We use the `print` command to actually see the plot.

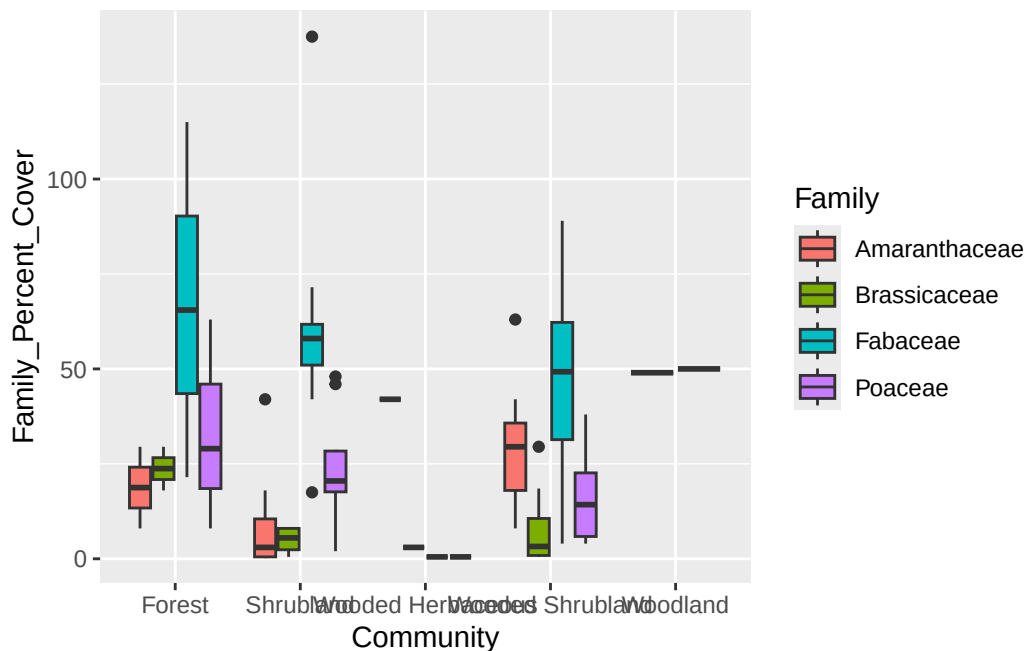
And here is what it looks like:



Telling the story

That's a fine start, but if we are interested in looking at differences among the families, we need an easy way of telling the families apart. We can do this in the `mapping` argument of the `ggplot` command by indicating that the shapes should be filled with different colors, according to value in the `Family` column:

```
# Boxplot of family-level data
cover_plot <- ggplot(data = family_data,
                     mapping = aes(x = Community, y = Family_Percent_Cover,
                                   fill = Family)) +
  geom_boxplot()
print(cover_plot)
```



Look at the Wooded Herbaceous and Woodland - they're just lines. Why? We can see the counts in each category using the `table` command:

```
table(family_data$Family, family_data$Community)
```

	Forest	Shrubland	Wooded Herbaceous	Wooded Shrubland	Woodland
Amaranthaceae	2	7	1	7	0
Brassicaceae	2	4	1	8	0
Fabaceae	3	8	1	8	1
Poaceae	3	8	1	8	1

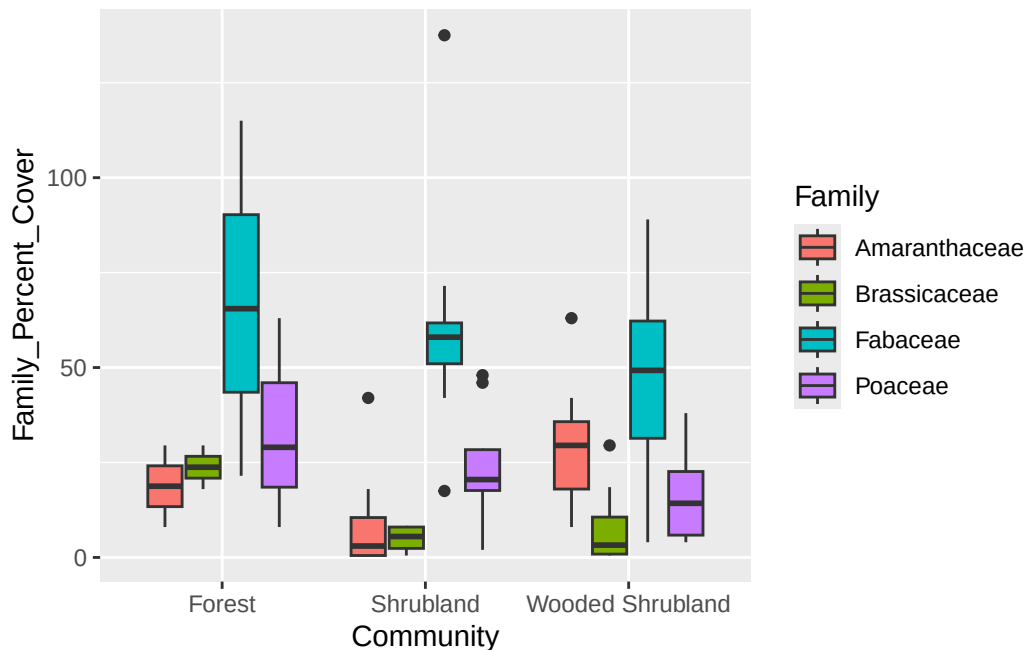
Since we only have those single plots for Wooded Herbaceous and Woodland, we should probably exclude them from our plot.

```
# Too few observations in two communities, so remove them
family_data <- family_data[!(family_data$Community %in% c("Wooded Herbaceous", "Woodland")),]

# Boxplot of family-level data
cover_plot <- ggplot(data = family_data,
  mapping = aes(x = Community, y = Family_Percent_Cover,
    fill = Family)) +
```



```
geom_boxplot()
print(cover_plot)
```

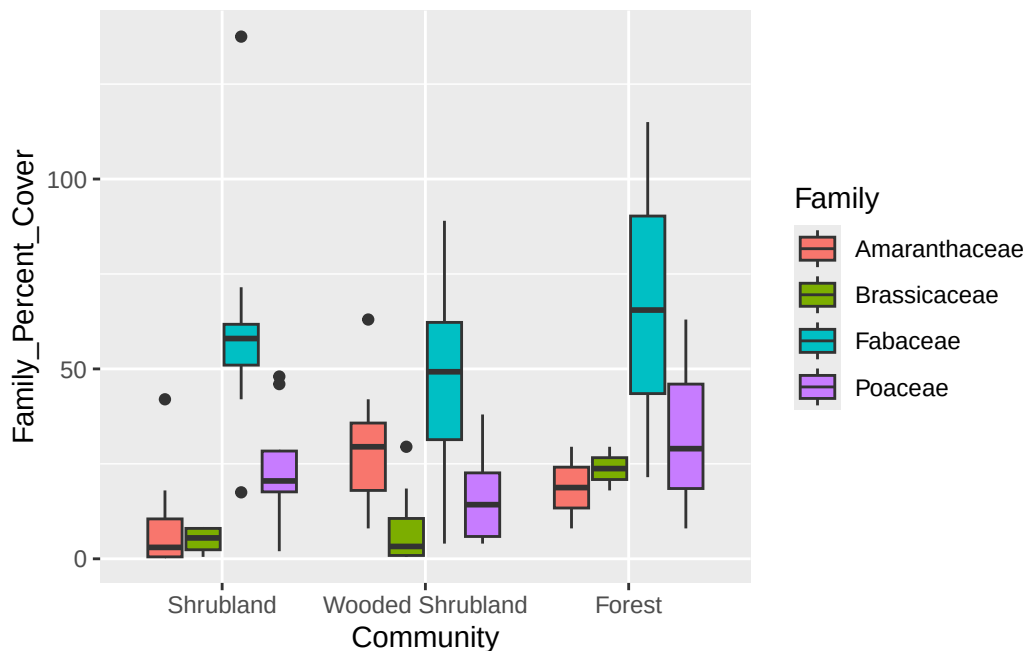


Now we only have those three communities. We would also like to rearrange our x-axis so the different categories of communities have an increasing number of trees, reading left to right. That is, we want the left-most community to be Shrubland, the middle community to be Wooded Shrubland, and the right-most community to be Forest. We do this by making a slight tweak to the data, where we “re-level” the categories in the `Community` column of our data:

```
# Re-order levels of Community so they plot in desired order
family_data$Community <- factor(family_data$Community,
                                levels = c("Shrubland",
                                             "Wooded Shrubland",
                                             "Forest"))

# Boxplot of family-level data
cover_plot <- ggplot(data = family_data,
                     mapping = aes(x = Community, y = Family_Percent_Cover,
                                    fill = Family)) +

  geom_boxplot()
print(cover_plot)
```



Finally, we can save the plot to a file using the `ggsave` function:

```
ggsave(plot = cover_plot, filename = "output/family-cover-plot.pdf")
```

This saves the file in the “output” folder with the name “family-cover-plot.pdf”.

Our final script would look like this:

```
# Plotting Tumacacori vegetation
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# 2019-09-06

# Libraries required for data wrangling and plotting
library(dplyr)
library(ggplot2)

plant_data <- read.csv(file = "data/tumacacori-vegetation.csv",
                      stringsAsFactors = TRUE)

# Do not want rows with missing data
plant_data <- na.omit(plant_data)
```

```

# Only focus on four families
families_keep <- c("Fabaceae", "Poaceae", "Brassicaceae", "Amaranthaceae")

# Create new data with only four families
subset_data <- plant_data[plant_data$Family %in% families_keep, ]

# Re-level data in the Family column
subset_data$Family <- factor(subset_data$Family)

# Calculate summary statistics for all species in subsetted data
cover_mean <- mean(subset_data$Percent_Cover)
cover_sd <- sd(subset_data$Percent_Cover)

# Calculate mean and standard deviation for mesquite alone
mesquite_mean <- mean(subset_data$Percent_Cover[subset_data$Common_Name == "velvet mesquite"])
mesquite_sd <- sd(subset_data$Percent_Cover[subset_data$Common_Name == "velvet mesquite"])

# Calculate mean and standard deviation for each species separately
summary_stats <- subset_data %>%
  # make a separate group of data for each Common_Name
  group_by(Common_Name) %>%
  # calculate the mean and standard deviation of Percent_Cover and store them
  #   in columns called "mean_cover" and "sd_cover" respectively
  summarize(mean_cover = mean(Percent_Cover),
             sd_cover = sd(Percent_Cover))

# Calculate family-level total percent cover for each plot separately
family_data <- subset_data %>%
  group_by(Plot_Code, Family, Community) %>%
  summarize(Family_Percent_Cover = sum(Percent_Cover))

# Too few observations in two communities, so remove them
family_data <- family_data[!(family_data$Community %in% c("Wooded Herbaceous", "Woodland")), ]

# Re-order levels of Community so they plot in desired order
family_data$Community <- factor(family_data$Community,
                               levels = c("Shrubland", "Wooded Shrubland", "Forest"))

# Boxplot of family-level data
cover_plot <- ggplot(data = family_data,
                     mapping = aes(x = Community, y = Family_Percent_Cover,
                                   fill = Family)) +

```

```
geom_boxplot()
print(cover_plot)

ggsave(plot = cover_plot, filename = "output/family-cover-plot.pdf")
```

Additional resources

- Official [ggplot documentation](#)
- A handy [cheatsheet for ggplot](#)
- A [PDF version](#) of this lesson